Lesson 6 — What Each Notebook Shows

Notebook guide, lesson 6 — Technologies for Artificial Intelligence

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A short guide to this lesson's notebooks. For each one: what it is for, the concepts in the order they appear, the numbers worth noticing, and **the two-minute version**, a summary to read before running it or to come back to afterwards. Every number here comes from the committed output of the notebook it belongs to.

Notebook 01 — `01_knn_and_the_curse`

k-nearest neighbours, and where distance stops working

What it is for. Every model so far learned a rule — coefficients, a boundary, a threshold. This one learns nothing at all, wins anyway, and is then broken on purpose.

The data. 1,200 pumps, two readings each, **735 faulty (61.3%)**, with the healthy ones in a central blob and the faulty ones surrounding it.

The concepts, in order.

1. **Look at it first**, and see that **no straight line separates the classes**. A pump can fail by running too slow *or* too fast.
2. **What that costs a linear model.** Majority baseline **0.613**; logistic regression **0.613 ± 0.000**. Not slightly beaten — it learned *nothing*, and predicts "faulty" for every pump.
3. **k-NN written out in eight lines**, agreeing with scikit-learn on **100% of the test points**, accuracy 0.937. Then **one prediction, drawn**: a new pump at 46 Hz and 5.8 bar, its five

nearest stored pumps, and their vote, four faulty to one healthy - the healthy one a flipped label, by the rule that generated the data.

4. **Choosing** k, the bias-variance dial made visible:

k	training (each fold's 960 stored pumps)	cross-validated
1	**1.000**	0.912
5	0.954	**0.944**
51	0.941	0.933
401	0.713	0.708

   At $k = 1$ the training accuracy is exactly 1.000 **by construction** — every point is its own nearest neighbour — which is the clearest demonstration that a training score can measure nothing at all.

5. **The three boundaries.** Ragged with islands at $k = 1$, a clean ellipse at $k = 15$, and swollen past the true envelope at $k = 401$.

6. **Now break it.** Add columns of pure noise, leaving both real readings untouched; the experiment is drawn first, as the table k-NN receives with ten noise columns:

noise columns	accuracy	above baseline
0	0.938	+0.325
10	0.762	+0.149
50	0.602	**−0.011**
100	0.578	−0.035

   The signal never left, and ten useless columns cost eighteen points.

7. **Why: geometry, not k-NN.** The ratio of nearest to farthest distance — **0.016** in 2 dimensions, 0.263 at 10, **0.701 at 100**, 0.855 at 500. Once every point is about as far as every other, "nearest" stops being a statement.

8. **What it costs to use.** No training at all, and $O(mn)$ per prediction — plus the dataset itself has to travel with the model.

The two-minute version. *Two readings per pump, healthy in the middle and faulty all around, so no line works: logistic regression scores exactly the base rate, 0.613. k-NN, which learns nothing, scores 0.944 with k = 5. At k = 1 the training accuracy is 1.000 by construction and the honest score is 0.912 — a perfect training score meaning nothing at all. Then we break it: adding ten columns of pure noise, with both real readings untouched, costs eighteen points of accuracy, and by fifty columns k-NN is worse than always guessing "faulty". The mechanism is geometric — in two dimensions the nearest point is 2% as far as the farthest, in a hundred it is 70%.*

Notebook 02 — `02_naive_bayes`

Naive Bayes, and the assumption it is named for

What it is for. k-NN made no assumptions and paid for it in dimensions. This is the opposite bargain: one very strong assumption, in exchange for training in a single pass and working in thousands of dimensions. The notebook shows it succeeding, then failing **one step away** — and explains both.

The concepts, in order.

1. **Bayes' rule, and where the difficulty is.** The prior and the denominator are easy; the joint density $p(x \mid y = c)$ is the problem.
2. **The naive assumption** — given the class, the features are independent — which replaces one n-dimensional estimate with n one-dimensional ones.
3. **On the pumps it does very well: 0.933 ± 0.010**, one point behind k-NN and a long way ahead of logistic regression's 0.613.
4. **And then the check that makes it a lesson rather than a result.** The assumption is about *within*-class dependence: overall −0.046, **within healthy −0.006, within faulty −0.049.** Uncorrelated — but not independent. Healthy pumps fill a disc and faulty ones a ring, and on a ring a pump far out in vibration must be central in pressure; squaring the distances from the design point shows it, **+0.168** healthy and **−0.420** faulty. **The assumption is false here**, and why it works anyway is item 6.
5. **What training actually computed.** Gaussian Naive Bayes puts a bell curve on every feature in every class, so training is priors plus a mean and a variance per feature per class — **ten numbers** on the pumps, one pass, no gradient descent. The table carries the surprise: the classes have **the same centre** (41.986 against 42.015 Hz) and different **widths** (sd 1.695 against 4.636 Hz; 0.219 against 0.576 bar). That is why logistic regression is stuck at the baseline and Naive Bayes is not. One pump taken apart term by term: at 48 Hz and 5.6 bar the log scores are −8.095 healthy and −4.144 faulty, **P(faulty) = 0.981**, matching `predict_proba`; at the design point, **P(healthy) = 0.820**. The positive log P(pressure), +0.598, is a density above 1, not an error.
6. **Wrong densities, nearly right boundary.** Walking out from the centre until the prediction flips puts the model's edge at 3.152 and 3.191 Hz, 0.413 and 0.411 bar, against a true envelope of 3.50 Hz and 0.45 bar: an axis-aligned ellipse of the true shape, about a tenth small. With shared centres, comparing a narrow bell to a wide one can only give such an ellipse — which is why the false assumption costs almost nothing: 0.933 against k-NN's 0.944, under the 0.96 noise ceiling.
7. **One step away, worse than guessing.** A second dataset where the pump is faulty when **exactly one** reading is high. Each sensor alone correlates +0.081 and −0.064 with the label — essentially nothing. The redrawn figure shows why: per sensor both classes have the same two humps (quadrants 273/278 healthy, 359/253 faulty), and the one bell per class Naive Bayes fits sits in the valley, nearly identical for the two classes (sensor A: centre −0.012 and +0.167, sd 1.090 and 1.112). The 0.18 between centres (0.14 on sensor B) is a sampling accident. Results: baseline 0.523, **Naive Bayes 0.404**, logistic regression 0.393, linear support vector machine (SVM) 0.606, k-NN 0.967, RBF SVM 0.972. k-NN and the RBF SVM prove the data is easily learnable: **the problem is the hypothesis class, not the data.**
8. **The assumption failing, measured.** Section 3's test on these sensors gives the mirror image: overall **−0.057**, but **+0.821 within healthy** and **−0.826 within faulty**. Independent-

looking overall, tightly tied inside each class — exactly what the model assumes away.

9. **It cannot even tell that it is lost.** Mean confidence **0.567 when right, 0.555 when wrong**, highest 0.673. Equally unsure either way, which is honest and useless: the reported probability carries no information about whether to trust the prediction.
10. **When to reach for it anyway.** Very high dimensions with little data — text classification being the canonical case.

The two-minute version. *Naive Bayes takes the opposite bargain from k-NN: one very strong assumption, and in exchange it trains in a single pass. On the pumps it scores 0.933, and rather than leave that as luck we check the assumption — within each class the two readings correlate -0.006 and -0.049, but squared distances correlate -0.420 among the faulty pumps: uncorrelated, not independent, so the assumption is false. Training is ten numbers, and they show why it works anyway: the two classes share a centre and differ only in width, so the boundary comes out an ellipse of the true shape, 3.17 Hz by 0.41 bar against 3.5 by 0.45. Then one step away: a dataset where the fault is "exactly one reading is high". Each sensor alone says nothing, and Naive Bayes only ever sees the sensors one at a time — inside each class the two correlate $+0.821$ and -0.826 — so it scores 0.404, below chance, while k-NN gets 0.967 on the same data. Structural, not a data shortage. And its confidence is 0.567 when right and 0.555 when wrong, so it cannot tell you when to doubt it.*

Notebook 03 — `03_svm_margins_and_kernels`

Support vector machines: the widest street, and a change of coordinates

What it is for. The third answer to the same question. k-NN remembered, Naive Bayes assumed; the SVM **changes the criterion** and then, with kernels, **changes the space**.

The concepts, in order.

1. **The widest street.** On separable data every separating line has zero error, so the error cannot choose between them. The SVM takes the widest slab — and of **80 points, 3 are support vectors**. Only a handful of points matter.

2. **Soft margins and the one knob.** Violations are allowed and charged for; C sets the price, and note the direction — large C means *less* tolerance and less regularisation.

3. **On the pumps, a straight line still fails.** Linear SVM **0.613 ± 0.000**, RBF **0.947 ± 0.005**. And the support-vector counts say why in a second language: **947 of 1,200 (79%)** for the linear kernel against **278 (23%)** for the RBF. A boundary pushing through a crowd is a boundary in the wrong space.

4. **The kernel trick.** Add a third coordinate — distance from the design point — and the disc-inside-a-ring becomes two layers a plane can separate. Then the idea that makes it famous: the dual depends on the data only through inner products, so a kernel computes those in the transformed space **without ever constructing the coordinates**. Checked by hand: for $\phi(x) = (x_1^2, \sqrt{2}x_1x_2, x_2^2)$, $u = (1, 2)$, $v = (3, 1)$, lifting gives $9 + 12 + 4 = 25$ and never lifting gives $(u \cdot v)^2 = 5^2 = 25$.

5. **Training and inference, step by step.** Training solves for one weight per pump, two at a time, no learning rate: **922** at weight 0, **268** at the ceiling $C = 1$, **10** on the edge —

278 support vectors — and $b = +1.105$. Inference on the 48 Hz, 5.6 bar pump: faulty votes **+37.455**, healthy **−36.677**, intercept +1.105, total **+1.883**, exactly `decision_function`: faulty. The 56 support vectors within half similarity give a net +1.141, the other 222 −0.363 — the neighbourhood decides, though the five nearest disagree. k-NN with a smooth similarity and learned weights.

6. **Gamma, C, and what they do to the boundary.** Similarity halves at 2.63, 1.18, 0.83 and 0.12 standard deviations for $\gamma = 0.1$, 0.5, 1 and 50:

setting	training (each fold's 960)	cross-validated
$\gamma = 0.1$, C = 1	0.934	0.929
$\gamma = 1$, C = 1	0.948	**0.944**
$\gamma = 50$, C = 1000	**0.996**	0.902

   Each knob alone barely hurts — $\gamma = 50$ with C = 1 scores 0.940, $\gamma = 1$ with C = 1000 0.948 — together they overfit.

   The best training score is the worst honest one — the signature lesson 5 taught them to read, now drawn as bubbles around individual pumps with flipped labels.

7. **The three methods, side by side**, with the noise ceiling at about 0.96 because 4% of labels are flipped: baseline 0.613, logistic regression 0.613, linear SVM 0.613, **Naive Bayes 0.933, k-NN 0.944, RBF SVM 0.947.** Three reach the ceiling by three different routes; two do not, and both are linear.

The two-minute version. *The SVM changes the criterion: among the many lines that separate the data, take the one with the widest slab around it — and on the example only three points of eighty decide it. On the pumps the linear SVM still scores the base rate, because the problem was never which line to choose, and 79% of the points end up as support vectors, which is what a boundary forced through a crowd looks like. Then the kernel trick: add distance-from-centre as a third coordinate and a plane separates the classes, and the kernel computes those inner products without ever building the coordinates — 25 both ways on a two-point example. Training finds one weight per pump, 922 of them zero; a prediction is the 278 support vectors voting by similarity, +1.883 for the 48 Hz pump. RBF scores 0.947 against a noise ceiling of 0.96. The last table is the lesson: three families reach the ceiling by three different mechanisms, the two linear models sit exactly on the baseline, and the gap is 0.334.*